Article

fp-tools: A Reproducible Platform for ATAC-seq Footprinting and Regulatory Motif Analysis

Yaoxiang Li [1] and Chunling Yi [1,*]

[1] Department of Oncology, Lombardi Comprehensive Cancer Center, Georgetown University Medical Center, Georgetown University, Washington, DC, USA; yl814@georgetown.edu (Y.L.); cy232@georgetown.edu (C.Y.)
* Correspondence: cy232@georgetown.edu (C.Y.)

Abstract

Transcription-factor (TF) footprinting from ATAC-seq provides a base-resolution view of regulatory factor occupancy, yet complete analyses often require multiple tools with different interfaces, signal conventions, and reproducibility practices. We present `fp-tools`, a Python package that integrates Tn5 bias correction, footprint calling, motif matching, differential footprint analysis, and aggregate visualization within a consistent command-line workflow. The same framework supports de novo motif-discovery preparation, pseudobulk aggregation from single-cell ATAC fragments, and replicate-aware reporting for condition comparisons. Using public bulk ATAC-seq replicates and 10x Genomics PBMC Multiome data, we show that `fp-tools` recovers biologically interpretable motif-centered aggregate patterns while preserving a reproducible analysis trail through versioned inputs, source tables, and regression-tested command outputs. `fp-tools` provides an integrated and auditable platform for ATAC-seq footprinting workflows and their motif-centered extensions.

Keywords: ATAC-seq; transcription factor footprinting; chromatin accessibility; motif analysis; pseudobulk ATAC-seq; replicate-aware reporting; reproducibility; command-line bioinformatics

1. Introduction

Chromatin accessibility profiling by the assay for transposase-accessible chromatin using sequencing (ATAC-seq) has become a standard readout of the active regulatory landscape of a cell [1]. Within accessible regions, sequence-specific transcription factors (TFs) protect the DNA they occupy from transposition. This creates local cut-site depletions, often called "footprints," with higher accessibility in the nearby flanks. Detecting and interpreting these footprints supports inference of TF occupancy, differential regulation between conditions, and the functional consequences of non-coding variation.

A mature but *fragmented* ecosystem of methods addresses pieces of this problem. ATAC-seq footprinting and differential-binding frameworks such as TOBIAS [2] and HINT-ATAC [3] correct enzymatic sequence bias and score footprints at a single TF-relevant scale. Multiscale and nucleosome-aware approaches, exemplified by the PRINT/scPrinter family [4], model accessibility structure across a range of footprint widths. Supervised TF-binding-site (TFBS) predictors such as maxATAC [5] and sequence-to-profile models such as ChromBPNet [6] learn occupancy from labeled data or raw sequence, and downstream interpretation tools and variant scorers translate models into motif and allele-level effects. Motif-discovery and motif-comparison tools, including MEME/STREME [7,8]

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and Tomtom [9], are also widely used after peak or footprint selection, but commonly run as separate steps with separate input and output conventions. Each tool typically ships its own interface, input conventions, and assumptions, so assembling a coherent, reproducible pipeline and *benchmarking* alternatives on common ground remains laborious and error-prone.

Beyond this tooling friction, footprint interpretation has practical limitations that make reproducibility and careful signal handling important. ATAC-seq footprinting depends on base-resolution cut-site depletion and can be confounded by enzyme sequence bias, local accessibility, motif strength, and factor-specific binding kinetics [10]. Intrinsic cleavage bias remains a major confounder in accessibility-based footprinting, motivating explicit bias correction and careful interpretation [11]. Deep sequence-to-profile models and multiscale methods can improve sensitivity for some questions, but they often require substantial compute and specialized training workflows. For many applied ATAC-seq projects, the immediate need is therefore a transparent, repeatable workflow that keeps footprint interpretation traceable while producing biologically informative reports within one software ecosystem.

Three practical gaps motivate this work. First, **signal clarity**: users need to know which signal representation is appropriate for footprint calling and how normalization changes motif-centered interpretation. Second, **interface fragmentation**: moving from bias correction to footprint calling, motif matching, differential footprint analysis, and visualization commonly spans several packages with incompatible I/O. Third, **reproducible biological reporting**: many workflows produce intermediate tracks and tables but do not provide a consistent path to replicate-aware summaries, motif-centered aggregate views, de novo motif follow-up, and single-cell pseudobulk analyses.

We address these gaps with `fp-tools`, a reproducible ATAC-seq footprinting platform that connects bias correction, footprint calling, motif matching, differential footprint analysis, and aggregate visualization through a single command-line workflow. The same framework integrates de novo motif-discovery preparation, pseudobulk ATAC aggregation, replicate-aware comparison, quantile normalization, and interactive HTML reports. This paper describes the implemented methods, public-data validation examples, and reproducibility framework that tie these workflows together.

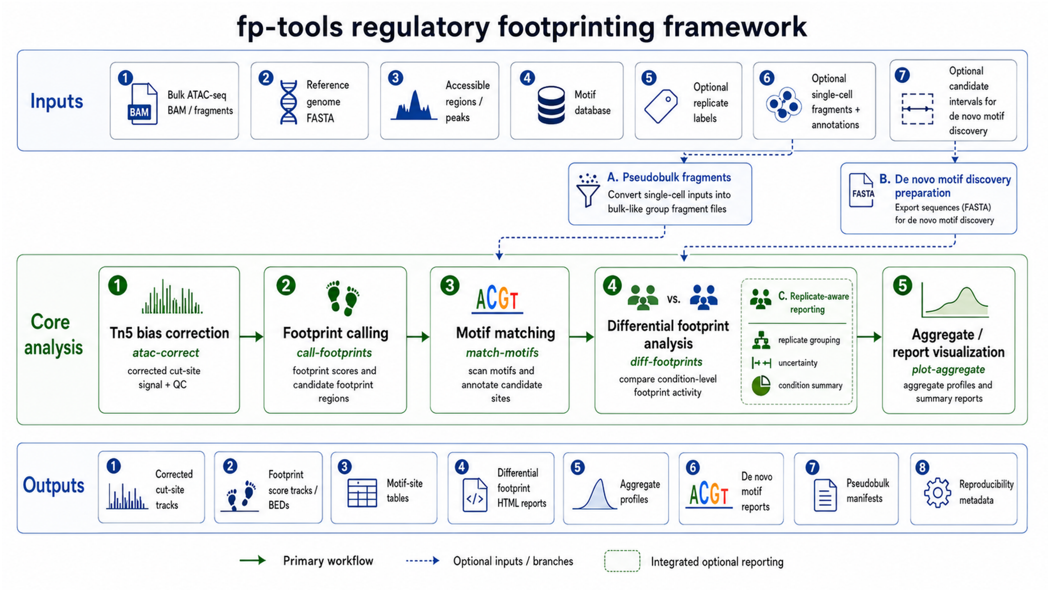

Figure 1. Overall `fp-tools` regulatory-footprinting framework. Bulk or single-cell ATAC-seq inputs enter an integrated core for Tn5 bias correction, footprint calling, motif matching, differential footprint analysis, and aggregate visualization. Integrated branches support single-cell pseudobulk fragment generation, de novo motif-discovery preparation from candidate intervals, and replicate-aware reporting within differential comparisons. Outputs include corrected cut-site tracks, footprint score tracks or candidate BEDs, motif-site tables, differential-footprint HTML reports, aggregate profiles, pseudobulk manifests, de novo motif reports, and reproducibility metadata.

2. Materials and Methods

2.1. Package Architecture and Core Workflows

`fp-tools` implements performance-critical signal operations with Cython-backed routines. The workflow has four method layers: Tn5 sequence-bias correction, footprint scoring over corrected signal, motif-anchored differential TF-binding analysis, and aggregate visualization around TFBS or region sets. These layers share common region, signal, motif, and sample-label conventions so intermediate files can be passed through a complete analysis without format-specific glue code. YAML configuration files provide a reproducible way to save and rerun the same analysis without changing the underlying methods.

Following the signal model popularized by TOBIAS [2], `fp-tools` treats base-resolution, bias-corrected cut-site profiles as the primary footprinting signal. For footprint calling, let x_i denote corrected Tn5 cut-site signal around candidate position p. The default footprint score is a center-versus-flank contrast,

$$F(p) = \frac{1}{|L| + |R|} \left(\sum_{i \in L} x_i + \sum_{i \in R} x_i \right) - \frac{1}{|C|} \sum_{i \in C} x_i,$$

where C is the protected center and L and R are flanking windows. Larger values therefore correspond to the expected footprint pattern: fewer cuts at the putative bound site and more cuts in nearby accessible flanks.

2.2. De Novo Motif Discovery Preparation

The de novo motif-discovery workflow begins from candidate footprint intervals, which may be produced by footprint calling or supplied by the user. For each candidate interval j with center p_j, strand a_j, and flank size h, the workflow exports a candidate-centered sequence

$$S_j = G[p_j - h, p_j + h],$$

reverse-complementing the sequence when strand information indicates the negative strand. The resulting FASTA file is accompanied by a reproducibility record that stores the input intervals, genome build, flank width, random seed, external tool versions, and the exact MEME/STREME [7,8] and Tomtom [9] commands. STREME was designed for accurate and versatile motif discovery from enriched sequence sets [8]. The same entry point can be used in two modes. In de novo-only mode, discovered motifs are reported as an independent motif set. In database-plus-de-novo mode, discovered motifs are matched to a known database such as JASPAR2026 and can be appended to the known motif set for downstream differential footprint analysis.

2.3. Pseudobulk Fragment Aggregation

The pseudobulk workflow groups single-cell ATAC fragments by a metadata column, such as cell type, cluster, donor, or treatment. Let F_b be the set of fragments assigned to barcode b, and let $g(b)$ be its group label. The fragment set for a pseudobulk group k is

$$F_k = \bigcup_{b:g(b)=k} F_b.$$

Groups that pass user-defined cell-count and fragment-count filters are written as bulk-like fragment files with manifest records. When genome sizes are supplied, `fp-tools` can also emit CPM-normalized cut-site bigWigs,

$$y_{k,t} = 10^6 \frac{c_{k,t}}{\sum_u c_{k,u}},$$

where $c_{k,t}$ is the cut-site count for group k at genomic position t. These tracks can be used directly by the same motif-centered aggregate plotting workflow used for bulk ATAC-seq data.

2.4. Replicate-aware Differential Footprint Analysis and Quantile Normalization

Replicate-aware analysis is integrated directly into the differential footprint comparison workflow. The differential comparison is a reporting-level generalization of the TOBIAS/BINDetect motif-associated footprint framework [2]: the same motif-centered footprint-score logic is retained, while repeated condition labels, replicate-level summaries, and optional quantile normalization are added around the comparison. Repeated condition names, for example `-cond-names Bcell Bcell Tcell Tcell`, define biological replicate groups while preserving the original differential-footprint output structure. Let $x_{c,r,i}$ be the footprint score at site or background position i for replicate r of condition c.

With `-normalization sample-quantile`, each replicate is normalized before condition averaging. For sample s, `fp-tools` computes empirical positive score quantiles $q_s(p)$ and the reference quantile $q_*(p) = S^{-1}\sum_s q_s(p)$ across the S input samples, then fits a smooth multiplicative factor $g_s(x)$ to the ratio $q_*(p)/q_s(p)$. Each score is transformed as

$$\tilde{x}_{s,i} = \max\{0,\, x_{s,i} g_s(x_{s,i})\}.$$

The `condition-quantile` mode first averages replicate background distributions within each condition, fits one factor $g_c(x)$ per condition, and applies that condition-level factor to all replicates in the condition. `none` disables quantile normalization. The same normalization helper is used by aggregate plotting, so differential comparison and motif-centered visualization are normalized consistently.

After normalization, condition-level scores are summarized as

$$\bar{x}_{c,i} = \frac{1}{n_c}\sum_{r=1}^{n_c}\tilde{x}_{c,r,i}, \qquad s_{c,i} = \sqrt{\frac{1}{n_c-1}\sum_{r=1}^{n_c}(\tilde{x}_{c,r,i}-\bar{x}_{c,i})^2}.$$

For a comparison between conditions c_1 and c_2, `fp-tools` reports a footprint-score difference and a stabilized log fold change,

$$\Delta_i = \bar{x}_{c_1,i} - \bar{x}_{c_2,i}, \qquad L_i = \log_2\frac{\bar{x}_{c_1,i}+\epsilon}{\bar{x}_{c_2,i}+\epsilon},$$

where ϵ is estimated from the background signal unless supplied explicitly. When both conditions have replicate support, approximate standard errors are reported as

$$\mathrm{SE}(\Delta) = \sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}},$$

$$\mathrm{SE}(L) \approx \frac{1}{\ln 2}\sqrt{\frac{s_1^2}{n_1(\mu_1+\epsilon)^2}+\frac{s_2^2}{n_2(\mu_2+\epsilon)^2}}.$$

For each comparison, `fp-tools` reports the standard `change` and `pvalue` columns together with a Benjamini-Hochberg-adjusted `qvalue_bh` column and a `significant_fdr05` flag. The `highlighted` column is retained as a report-selection and visualization flag, not as a formal multiple-testing-corrected significance label. Additional replicate-oriented columns include replicate counts, condition score SDs, mean Δ, mean L, and standard-error summaries.

2.5. Public ATAC Processing and Reproducibility Provenance

The bulk replicate demonstration uses true biological replicates from the Buenrostro et al. ATAC-seq study [1]: GM12878/B-cell samples SRR891268 and SRR891269, and day-1 CD4 T-cell samples SRR891275 and SRR891276. Reads were trimmed with fastp [12], aligned with Bowtie2 [13], processed with SAMtools [14] and BEDTools [15], and peaks were called with MACS3 using the MACS model-based peak-calling framework [16]. Problematic genomic regions were filtered using the ENCODE blacklist [17]. Replicate peaks were merged into a common peak universe, bias-corrected, footprint-called, and analyzed by differential footprint comparison with and without sample-level quantile normalization. The analysis used JASPAR2026 CORE vertebrates non-redundant motifs for known-motif scanning. Exact command-line parameters and software versions are recorded in source TSV files in the manuscript `tables` directory and summarized in Tables S1 and S2.

2.6. Public Data, Benchmark Tiers, and Metrics

Benchmarks are organized as versioned TSV manifests recording source, benchmark tier, cell type, donor, TF, assay, accessions, assembly, output type, format, URL, checksum, status, local path, split, and notes. A discovery script queries the ENCODE REST API for released human GRCh38 ATAC-seq and matched TF ChIP-seq/CUT&RUN; a resumable downloader writes per-benchmark download reports with checksums and local paths. Raw and processed public data, and large benchmark outputs, are kept out of version control; scripts, manifests, schemas, small reference inputs, and figure code are tracked.

Benchmark design followed published guidelines for fair and reproducible computational method benchmarking [18,19]. The benchmark is organized as a tiered design (Table 1): a small-reference regression tier; a bulk matched tier (ENCODE bulk ATAC vs. matched TF ChIP/CUT&RUN) for ATAC-seq footprinting benchmarking; a depth-robustness tier over downsampled ATAC; a de novo motif-preparation tier; and a real-data pseudobulk demonstration tier. For binary TF-binding labels we compute AUROC and AUPRC globally and per TF-cell task, recall at 1%, 5%, and 10% FDR, precision at fixed recall, calibration curves with expected and maximum calibration error, Brier score for probability models, per-TF macro averages, Wilcoxon signed-rank tests for paired method comparisons, and stratified bootstrap confidence intervals for key deltas. Expected calibration error partitions predictions into B equal-width probability bins and averages the gap between confidence and accuracy,

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{n} |\mathrm{acc}(b) - \mathrm{conf}(b)|,$$

with maximum calibration error defined analogously as the worst-bin gap. Metric, calibration, and bootstrap computations are implemented as unit-tested scripts and combined by a benchmark pipeline runner that also emits PDF/SVG/PNG figure panels.

Table 1. Tiered benchmark design. Large data and outputs are untracked; only manifests, scripts, and compact reference results are committed.

Tier	Purpose	Primary outputs
Small reference	Installation and regression checks on compact inputs	command success, output existence, stable summaries
Bulk matched	Main ATAC-seq footprinting benchmark (ATAC vs. matched TF ChIP/CUT&RUN)	TFBS labels, predictions, AUROC/AUPRC, reports
Depth robustness	Shallow-depth sensitivity (50/25/10/5%)	performance-vs-depth and calibration curves
De novo motif prep	Candidate FASTA and motif-discovery validation	exported FASTA, command plans, motif summaries
Pseudobulk	Single-cell extension and real-data demonstration	10x PBMC cell-type pseudobulks, TF aggregate profiles

2.7. Reproducibility Engineering

The package ships a unit-test suite covering numerical helpers, CLI behavior, and deterministic golden CLI regressions for footprint calling, differential footprint analysis, and aggregate plotting, with an opt-in slow regression for bias correction. A continuous-integration workflow runs the suite on a clean checkout using compact reference inputs, builds source and wheel artifacts, and verifies the installed console scripts. The repository includes `environment.yml`, a Dockerfile, release metadata, and Make targets for test, build, and manuscript-smoke workflows; these choices follow FAIR-oriented research-software principles and established bioinformatics environment practices [20–22]. Detailed reviewer instructions are kept in `docs/reproduce-paper.md`. Every benchmark result is expected to store random seeds, tool versions, command lines, and the resolved manifest. Figures are saved as vector PDF/SVG plus 300-dpi PNG previews, with source plotting tables retained for auditability.

3. Results

3.1. A Unified, Reproducible Footprinting Platform

`fp-tools` consolidates the core ATAC-seq footprinting steps into a single, consistent workflow with shared region/signal conventions, so a complete analysis, including bias correction, footprint calling, differential footprint analysis, and aggregate visualization, runs without crossing package boundaries or reformatting intermediates. YAML configuration records the same analysis in a machine-readable form for reruns and batch processing (Figure 1).

3.2. Interactive Interfaces and Report Review

In addition to direct command-line execution, `fp-tools` provides browser-based interfaces for configuring analyses and reviewing results. The GUI writes reusable YAML configurations that remain compatible with command-line execution. Differential-footprint and aggregate-batch workflows also produce self-contained HTML reports with embedded compressed data, searchable motif controls, editable group colors, synchronized aggregate profiles, and SVG export for publication-ready editing (Figure 2).

Figure 2. Usability-oriented `fp-tools` interfaces. (A) Browser GUI for configuring footprinting workflows and saving runnable YAML configurations. (B) Self-contained differential-footprint HTML report from the Buenrostro B-cell versus T-cell replicate comparison, combining differential evidence, selected motif metadata, motif logo display, and aggregate footprint profiles. (C) Aggregate-batch HTML report generated from the same embedded aggregate payload, allowing multi-sample and multi-TF aggregate review with searchable motif selection, group color controls, configurable panel layouts, and editable SVG export.

3.3. Regression-tested Workflow Outputs

The footprinting workflow is covered by deterministic regression tests that pin summary outputs for footprint calling, differential footprint analysis, and aggregate plotting. Public entry-point checks and YAML dry-runs verify that the command surface remains stable. These tests provide a reproducible contract for core analyses while allowing integrated modules to share the same input and output conventions.

3.4. De Novo Motif Discovery Complements Database-based Motif Analysis

The de novo motif-preparation path exports candidate-centered FASTA files, constructs reproducible MEME/DREME/STREME and Tomtom analysis records, and summarizes motif results into TSV/HTML reports with inline consensus logos. This keeps external motif discovery reproducible while relying on established motif-discovery algorithms. The workflow can be used alone to discover recurring patterns in candidate intervals, or after known-motif scanning to find additional motifs not represented in the selected database.

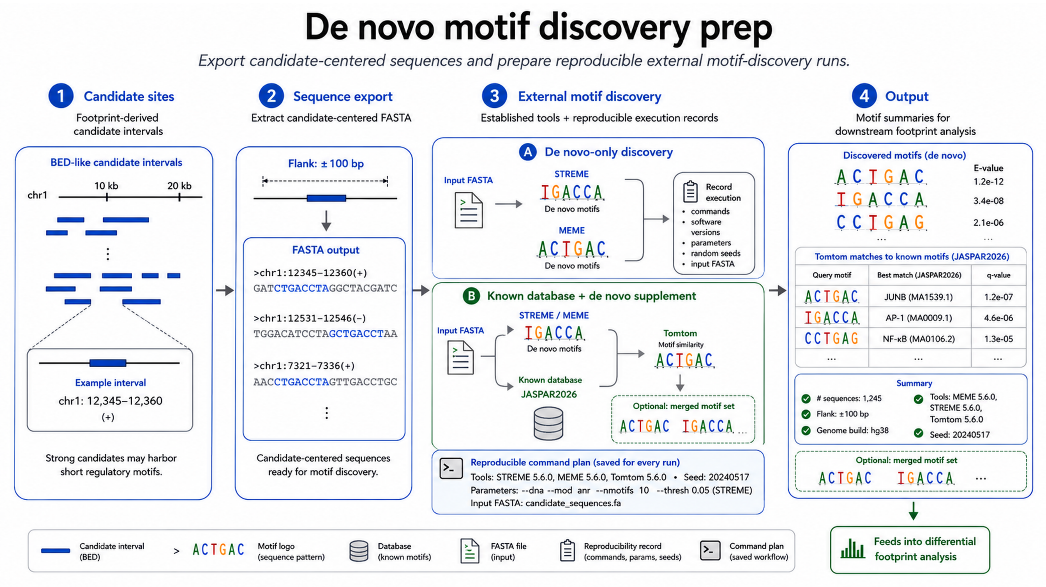

Figure 3. Generic de novo motif-discovery preparation workflow. Candidate footprint intervals are converted into candidate-centered FASTA sequences, external motif-discovery tools such as STREME or MEME are run with recorded parameters and software versions, and discovered motifs are summarized alone or matched back to known motif databases such as JASPAR2026. The same output can be used as a de novo result or as a supplemental motif set for downstream differential footprint analysis.

We therefore evaluated whether candidate-derived motifs add information beyond a fixed database using the same 2-vs-2 Buenrostro B-cell versus T-cell ATAC-seq comparison used for replicate-aware reporting. Candidate footprints were called from bias-corrected cut-site tracks, B-cell-enriched and T-cell-enriched candidate FASTA files were exported with 75 bp flanks, and STREME 5.5.9 was run in both contrast directions. The B-cell-enriched candidate set returned BATF-like and Irf1-like discovered motifs, whereas the T-cell-enriched candidate set returned IKZF2-like and ZNF143-like examples. These discovered motifs were then used in two modes. In de novo-only mode, the candidate-derived motifs were tested as their own motif set. In supplement mode, the same motifs were appended to a known motif database before running the sample-quantile-normalized differential-footprint analysis.

The full JASPAR2026 vertebrate non-redundant set contained 1019 motifs, while the de novo procedure contributed 16 candidate-derived motifs. In de novo-only mode, several discovered motifs had footprint-like center depletion in the embedded aggregate profiles; these aggregate examples were selected by negative center-versus-flank scores rather than by database annotation. In supplement mode, the discovered motifs were appended to known motif sets before the same motif-aware differential footprint analysis. To create a sensitivity test, we also constructed a restricted JASPAR2026 set with 860 motifs and reran the comparison with and without the 16 de novo motifs. The restricted database yielded 97 highlighted differential motif families; adding the 16 de novo motifs increased this to

101 highlighted families, including 6 de novo-derived highlighted motifs. This sensitivity test illustrates the intended use case: when a chosen motif database is incomplete or conservative, candidate-derived motifs can be evaluated on their own and can also be carried forward into the same differential footprint report as database-supported motifs (Figure 4).

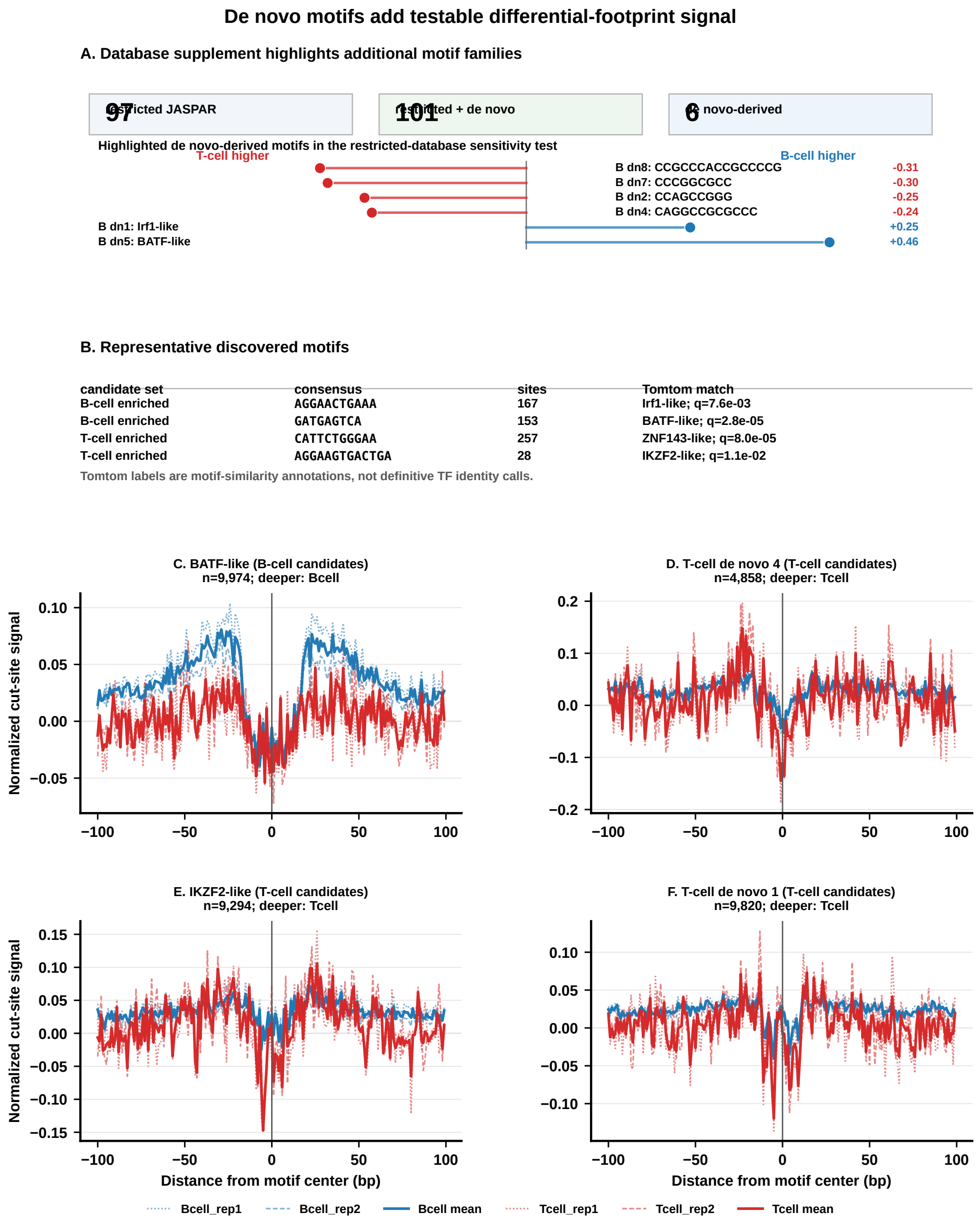

Figure 4. De novo motif validation in de novo-only and database-supplement modes. Candidate footprints from the Buenrostro B-cell versus T-cell replicate comparison were exported to FASTA and analyzed with STREME in B-cell-enriched and T-cell-enriched contrast directions. Panel A summarizes the restricted-database sensitivity test: adding 16 candidate-derived motifs increased the highlighted motif-family count from 97 to 101 and contributed 6 de novo-derived highlighted motifs. Panel B lists representative discovered motifs, including BATF-like and Irf1-like motifs from B-cell-enriched candidates and IKZF2-like and ZNF143-like motifs from T-cell-enriched candidates. Panels C–F show motif-centered aggregate profiles from the de novo-only analysis. Tomtom labels indicate motif similarity to database entries. Thin aggregate lines show individual biological replicates and thick lines show condition means; full displayed values are provided in the figure source table.

3.5. Replicate-aware Differential Footprint Analysis and Normalization

The replicate-aware workflow summarizes differential footprint results with effect sizes, replicate support, and uncertainty estimates while keeping the comparison anchored to a common peak and motif-site universe. In the Buenrostro example, B-cell and T-cell replicates are compared after motif scanning on the merged peak set, so no additional peak alignment step is performed at the reporting stage (Figure 5).

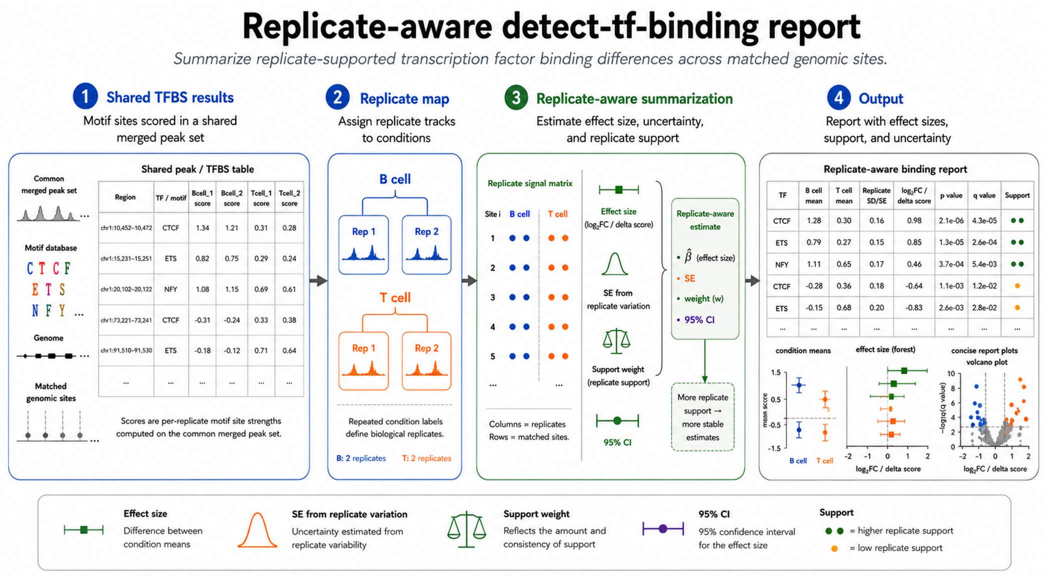

Figure 5. Example replicate-aware differential-footprint workflow for a B-cell versus T-cell comparison. Differential footprint results and a replicate map are combined to summarize effect size, replicate support, approximate uncertainty, and selected aggregate profiles. The workflow assumes that compared samples have already been mapped to a shared peak or motif-site coordinate system.

Replicate-aware differential footprint analysis adds an interpretable layer to condition comparison: repeated condition names define biological replicate groups, condition means are reported alongside individual replicate profiles, and sample-level quantile normalization applies the same distribution-matching step before replicate group averages are compared. To show the effect on real data, we used the B-cell versus T-cell 2-vs-2 Buenrostro comparison and extracted both the volcano summaries and motif-centered aggregate profiles from the embedded `diff-footprints` HTML payloads. The no-normalization run represents the TOBIAS-style baseline without an additional sample-level quantile-normalization step, whereas the sample-quantile run applies distribution matching before condition-level comparison and aggregate plotting. In this example, normalization reduced the positive distributional offset of motif-level differential footprint scores while preserving the condition-level aggregate profiles (Figure 6).

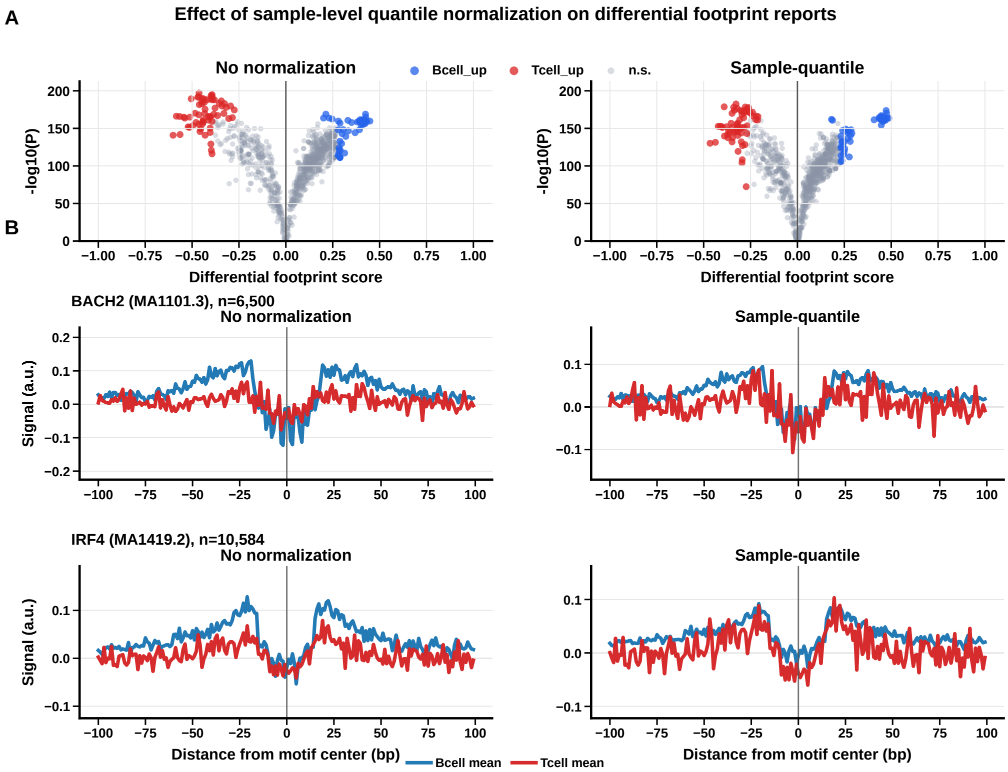

Figure 6. Effect of sample-level quantile normalization on the Buenrostro B-cell versus T-cell 2-vs-2 differential footprint report. (A) Volcano plots extracted from the no-normalization and sample-quantile `diff-footprints` HTML reports. The no-normalization comparison is the TOBIAS-style differential footprint baseline without additional quantile normalization. Sample-quantile normalization reduced the positive score offset (median differential footprint score 0.0998 to 0.0671) while retaining similar highlighted motif counts (55 to 53 B-cell-up motifs and 63 to 61 T-cell-up motifs). (B) BACH2 and IRF4 condition-mean aggregate profiles from the same report payloads. The aggregate y-axis is shown as signal in arbitrary units: corrected cut-site signal for no normalization and quantile-scaled corrected cut-site signal for sample-quantile normalization. A lower center relative to local flanks indicates stronger TF protection.

As a biological consistency check for the same 2-vs-2 Buenrostro replicate comparison, we inspected aggregate corrected cut-site profiles for selected motifs with visible footprint-like center depletion. BACH2 showed a B/GM12878-biased pattern in this dataset while remaining biologically non-exclusive because it is also reused in T-cell states. Conversely, JUNB and ATF7 showed stronger T-cell aggregate depletion; JUNB is an AP-1 family regulator of T-cell differentiation [23], and ATF7 is included here as a strong T-cell-deeper aggregate example rather than a lineage-specific marker. We also include IRF4 because TCR signaling can drive graded IRF4 expression in T cells [24], although its aggregate depletion is modest in this dataset. We treat these panels as an illustrative sanity check rather than a marker-classification benchmark; a more negative center-versus-flank profile indicates stronger footprint-like protection at motif centers (Figure 7).

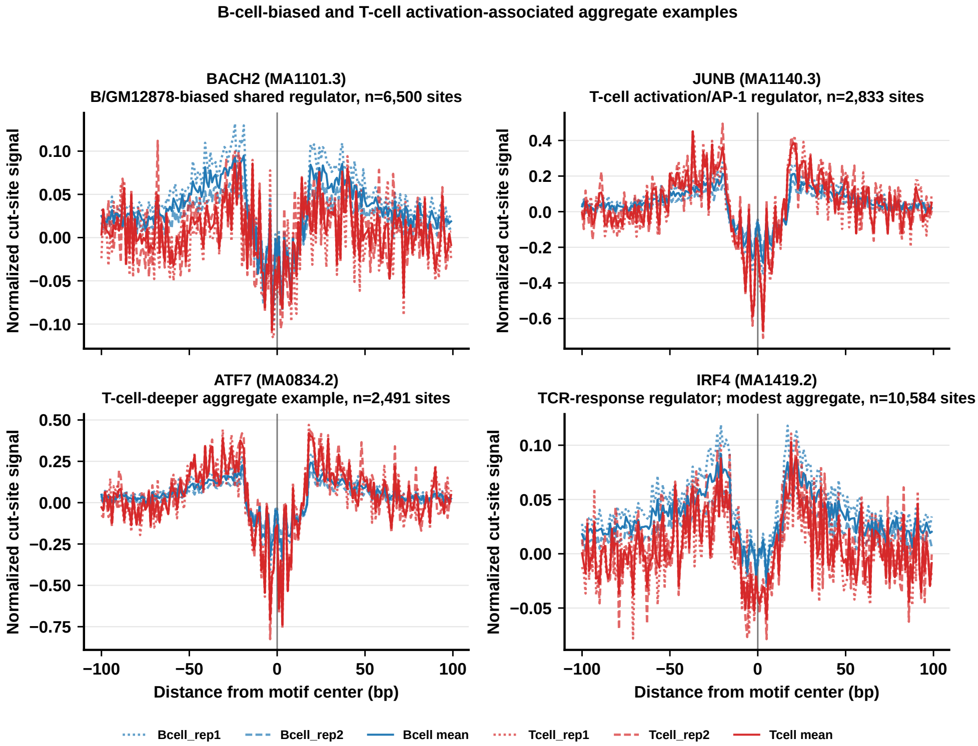

Figure 7. B-cell-biased and T-cell activation-associated aggregate examples from the B-cell versus T-cell replicate comparison. Aggregate profiles were generated from the same sample-quantile-normalized corrected cut-site payload embedded in the `diff-footprints` HTML report. Individual biological replicates and condition means are shown with comparable line widths to emphasize replicate consistency. BACH2 provides the B/GM12878-biased shared-regulator example; JUNB and ATF7 show stronger T-cell aggregate depletion; IRF4 is included as a biologically motivated TCR-response regulator with a modest aggregate effect in this dataset.

3.6. Single-cell ATAC Fragments Produce Interpretable Pseudobulk Footprints

We evaluated pseudobulk aggregation using the public 10x Genomics PBMC granulocyte-sorted Multiome dataset [25]. The workflow used the official 10x ATAC fragments and GEX graph clusters, mapped 11,898 cells into broad immune pseudobulks, and retained seven groups after requiring at least 300 cells and 50,000 fragments: B cells, CD4 T cells, NK/cytotoxic T cells, CD14 monocytes, FCGR3A monocytes, dendritic cells, and a mixed myeloid group. The prepared figure inputs used chromosomes chr1–chr22 plus chrX. The kept pseudobulks contained 13.6 to 144.0 million counted fragments per group and were written as indexed fragment files plus CPM-normalized cut-site bigWigs. These outputs were readable by the same aggregate-plotting code used for bulk ATAC data. For single-cell chromatin context, pseudobulk grouping was considered relative to established ecosystems including ArchR and Signac, and motif-activity baselines such as chromVAR [26–28].

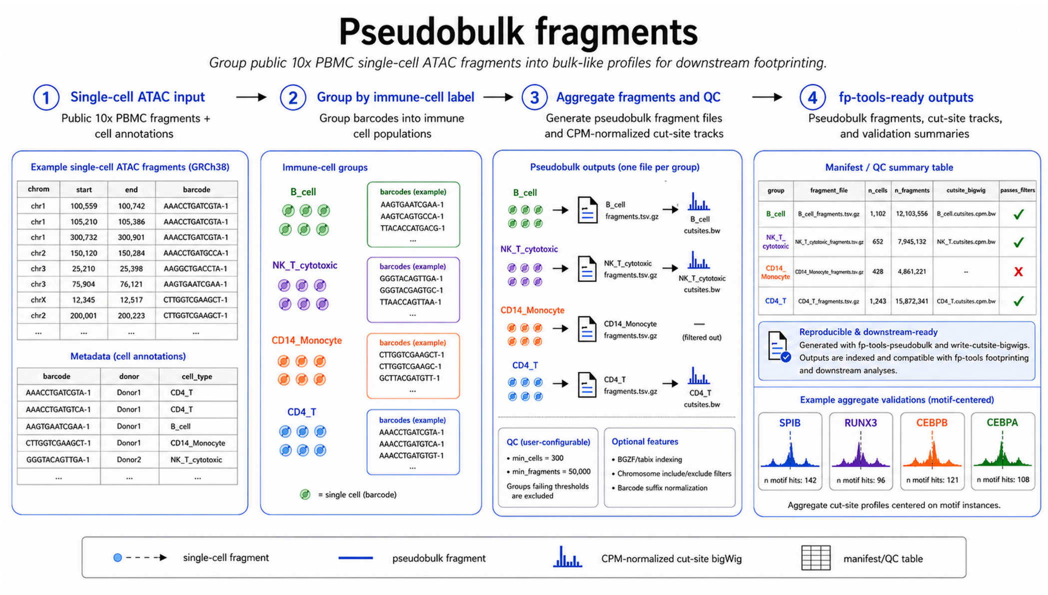

Figure 8. Example pseudobulk fragment workflow using public 10x PBMC single-cell ATAC data. Single-cell fragment records and cell annotations are grouped by broad immune-cell labels, converted into one pseudobulk fragment file per retained group, and summarized with QC fields, indexed outputs, and CPM-normalized cut-site bigWigs for downstream footprinting. The mini aggregate panels in the workflow illustrate motif-centered validation profiles from the same PBMC example, not a separate generic benchmark.

As a qualitative validation, we aggregated CPM-normalized pseudobulk cut-site signal in a $\pm$100 bp window around exact motif centers rather than around broad peak centers. JASPAR2026 vertebrate motifs were scanned in the 10x ATAC peak set, motif hits were grouped by TF family, and candidate immune regulators were screened using a predeclared B-cell, T/NK-cell, and myeloid marker list. To avoid artificial fixed site counts in the figure, all motif hits were counted and used for each displayed TF. We show RUNX3 as the T/NK-associated example and CEBPB as the myeloid-associated example from this screen, using 37,694 and 17,454 motif-centered sites, respectively. The fragment data and pseudobulk cut-site tracks were not modified; only the motif-centered site set used for aggregation changed. The resulting profiles show that single-cell fragments can be collapsed into group-level tracks and inspected with the same motif-centered aggregate workflow used for bulk ATAC data (Figure 9). The pseudobulk module can also emit pseudo-paired BAMs for downstream bias correction with zero read shifting.

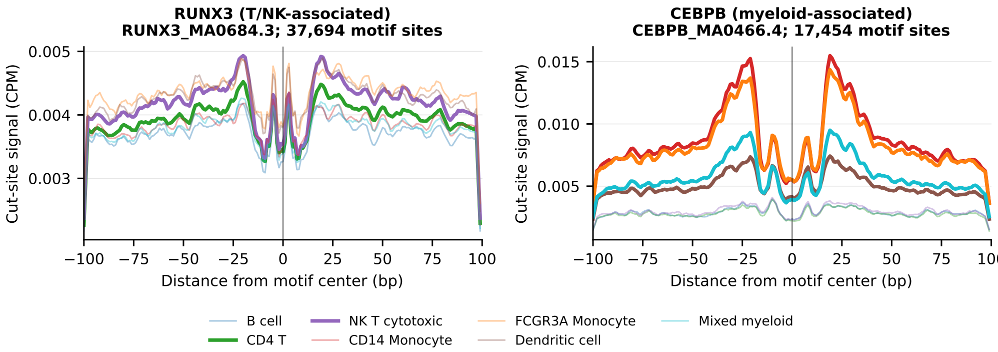

Figure 9. Motif-centered 10x PBMC pseudobulk cut-site signal aggregate profiles. Single-cell ATAC fragments from the public 10x PBMC Multiome dataset were grouped by broad immune-cell labels derived from official 10x gene-expression graph clusters, converted to CPM-normalized cut-site bigWigs, and aggregated in a ±100 bp window around exact JASPAR2026 motif centers scanned inside 10x ATAC peaks. RUNX3 is shown as a T/NK-associated example with 37,694 motif sites, and CEBPB is shown as a myeloid-associated example with 17,454 motif sites. The plotted value is direct CPM-normalized cut-site signal, not a derived protection score. Thin lines show all retained pseudobulk groups; thicker lines mark the expected pseudobulk groups for each lineage-associated TF. The source screen and natural motif-hit counts are reported in `figures/supp_pseudobulk_cutsite_signal_screen.tsv`. The figure demonstrates that the pseudobulk fragment workflow produces group-level signal tracks compatible with motif-centered aggregate visualization.

3.7. Benchmark and Figure Reproducibility

The tiered benchmark design (Table 1), metric scripts, manifest builders, schema validator, engineering-runtime recorder, and figure-generation scripts provide an executable path from public inputs to publication-ready outputs. These resources serve three purposes. First, they verify that the ATAC-seq footprinting workflow produces stable outputs on small fixtures and public-data subsets. Second, they provide matched-label inputs for bulk ATAC comparisons against TF ChIP-seq or CUT&RUN labels. Third, they support the manuscript demonstrations for de novo motif preparation, pseudobulk aggregation, replicate-aware normalization, and cut-site-resolution footprint calling.

Each benchmark run is expected to write a resolved manifest, software versions, parameters, and plotting tables next to the generated figures. This makes the figures auditable: a reader can inspect which samples, motifs, regions, and signal tracks were used, and the same tier can be rerun after code changes.

3.8. Cut-site Resolution Determines Whether Footprint Scores Are Interpretable

A footprint score is meaningful only when the input signal preserves the local cut-site depletion expected at bound TF motifs. We tested this point on K562 CTCF chromosome 4 peaks by taking each peak's best CTCF motif match and computing a simple flank-minus-center score from two signal representations (Figure 10). The smoothed fold-change-over-control coverage track is a good accessibility summary, but it is the wrong representation for this footprint statistic: the score is inverted and uninformative (AUROC = 0.24) because bound accessible sites often have high coverage at the motif region. In contrast, the same score computed from base-resolution Tn5 cut sites derived from the ATAC BAM recovers a footprint signal (AUROC = 0.68, 95% CI 0.65 to 0.70). This analysis clarifies the purpose of the upstream `fp-tools` correction and footprint-calling steps. The workflow is designed to operate on cut-site-resolution signal for footprint inference, while smoothed accessibility tracks are better treated as broader accessibility covariates or visualization tracks. For CTCF, the motif score remains the strongest single predictor; the cut-site footprint adds a distinct signal representation that becomes especially important when the question is condition-specific occupancy rather than motif presence alone.

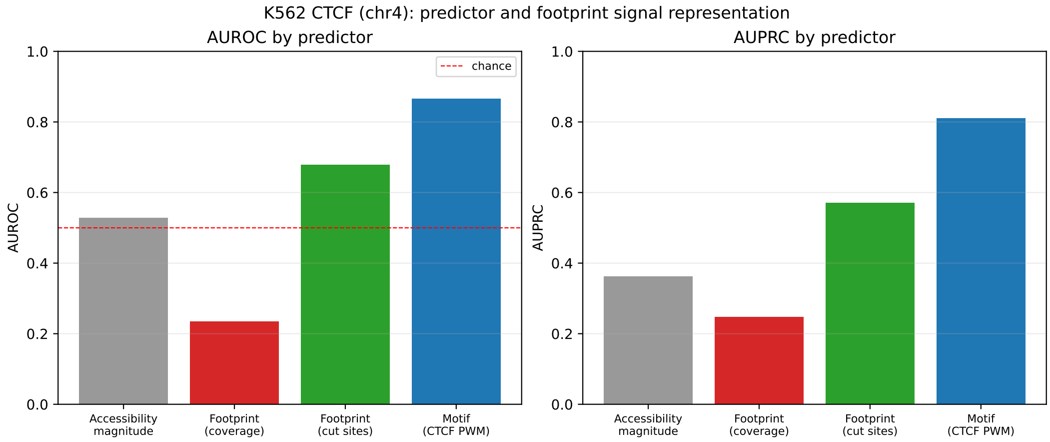

Figure 10. Predictor comparison for K562 CTCF (chromosome 4). A footprint-occupancy score is informative only at Tn5 cut-site resolution (green, AUROC 0.68); the same score on smoothed coverage is inverted and uninformative (red, 0.24). Raw accessibility (grey) is near chance and the CTCF motif (blue) is the single strongest predictor.

4. Discussion

`fp-tools` contributes a reproducible platform for ATAC-seq footprinting and motif-centered regulatory analysis. Its comparative strengths are interface consistency across bias correction, footprint calling, differential footprint analysis, and aggregate visualization; deterministic regression coverage; explicit handling of replicate-aware summaries; and integrated support for de novo motif preparation and pseudobulk aggregation using the same region, motif, and signal conventions (Table 2). Together, these properties address the fragmentation that often separates footprint scoring, motif interpretation, and report generation.

Table 2. Feature comparison across widely used ATAC-seq regulatory-analysis tools. ✓, native first-class support; ○, partial or indirect support; NA, not available. `fp-tools` is an integrated, reproducible platform that combines ATAC-seq footprinting with replicate-aware reporting, de novo motif-discovery preparation, and single-cell pseudobulk aggregation behind one command surface.

Feature	fp-tools	TOBIAS	HINT-ATAC	PRINT/ scPrinter	Chrom- BPNet	maxATAC
Bulk ATAC footprinting	✓	✓	✓	✓	○	NA
Tn5 bias correction	✓	✓	✓	✓	✓	NA
Footprint calling	✓	✓	✓	NA	NA	NA
Motif-aware differential footprint analysis	✓	✓	○	○	NA	NA
De novo motif-discovery preparation	✓	NA	NA	✓	✓	NA
scATAC / pseudobulk support	✓	○	○	✓	○	○
Replicate-aware reporting	✓	○	○	○	NA	NA
Visualization / reporting	✓	✓	○	✓	✓	○
YAML / batch execution	✓	NA	NA	NA	NA	NA

Deep sequence-to-profile models such as ChromBPNet provide powerful sequence-driven occupancy and variant-effect analyses [6]. Differential TF activity was also considered relative to orthogonal accessibility-based frameworks such as diffTF [29]. `fp-tools` addresses a complementary need: a lightweight, auditable ATAC-seq footprinting workflow that builds on the widely used TOBIAS-style footprinting model while integrating footprint calling, motif discovery, differential analysis, and reporting within one reproducible ecosystem. The replicate-aware summaries reported here use condition-level and replicate-level score variation to aid interpretation of differential footprint comparisons; they are designed for transparent reporting and visualization rather than as a replacement for specialized hierarchical models.

Limitations remain. The present benchmark evaluates discrimination within each cell–TF task under chromosome-held-out cross-validation; whole-genome cross-cell-line transfer (training on some cell lines and testing on others) and direct head-to-head runs of external tools such as TOBIAS, HINT-ATAC, and ChromBPNet on identical locus sets will require larger external-tool benchmark runs. Large public-benchmark inputs depend on external data acquisition and are therefore regenerated from committed manifests rather than stored directly in the repository. Natural next steps include larger external-tool comparisons, stronger per-replicate statistical models, and additional public validation datasets.

5. Conclusions

We presented `fp-tools`, a reproducible platform that unifies ATAC-seq footprinting with motif-discovery preparation, pseudobulk aggregation, and replicate-aware reporting. By pairing test-covered workflows with public-data manifests, source tables, and reproducible figure generation, the package supports standardized and benchmarkable regulatory-genomics analysis. Future work will add larger whole-genome transfer studies, direct external-tool comparisons, multiscale and nucleosome-aware validation, and stronger hierarchical differential-binding validation.

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Informed Consent Statement: Not applicable.

Data Availability Statement: `fp-tools` is openly available at https://github.com/oncologylab/fp-tools. All benchmark manifests, schemas, analysis scripts, and figure generators are included in the repository. Public input datasets are obtained from the ENCODE Project [30], the 10x Genomics PBMC Multiome example [25], and motif databases (e.g., JASPAR [31]) via the included discovery and download scripts. Manifests, scripts, small result tables, source tables, and figures are stored in the main repository. Reviewer reproduction instructions are provided in `docs/reproduce-paper.md`, with Conda and Docker execution files in `environment.yml` and the repository Dockerfile. Large reusable example-data files are distributed separately through the public `oncologylab/fp-tools-data` repository release assets, and full raw public inputs are regenerated from committed manifests rather than stored in version control.

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Conflicts of Interest: The author declares no conflicts of interest.

Appendix

Table S1. Primary analysis parameters used for the public ATAC and pseudobulk workflows. The complete tab-separated source table is stored as `analysis_parameters.tsv`.

Step	Parameters
FASTQ preprocessing	Paired-end adapter detection with `fastp`; HTML and JSON QC reports written per sample.
Alignment and BAM filtering	`bowtie2 -very-sensitive -X 2000` against hg38; retain proper pairs with `samtools view -f 2 -F 2828 -q 30`; sort, `fixmate -m`, coordinate sort, `markdup -r`, index, flagstat, and idxstats.
Blacklist filtering	Exclude mitochondrial and nonstandard contigs and subtract hg38 blacklist v2 regions.
Peak calling and union peaks	MACS3 paired-end peak calling with human genome size, duplicate retention, and q-value 0.01; replicate peaks are sorted and merged into one analysis peak set.
Bias correction and footprint calling	`atac-correct` with merged peaks and blacklist using default Tn5-bias settings; `call-footprints -score footprint` with default footprint/flank windows.
Differential footprint analysis	Four replicate footprint tracks analyzed with condition labels Bcell, Bcell, Tcell, Tcell; JASPAR2026 vertebrate non-redundant motifs; replicate report enabled; runs performed with no normalization and with sample-level quantile normalization.
Pseudobulk PBMC workflow	10x PBMC Multiome fragments grouped by broad immune labels; groups retained with at least 300 cells and 50,000 fragments; CPM-normalized cut-site bigWigs generated for aggregate profiles.

Table S2. Software and resource versions used for manuscript analyses. The complete source table is stored as `software_versions.tsv`.

Tool or resource	Version	Role
`fp-tools`	0.1.7	Footprinting, motif analysis, pseudobulk, and plotting utilities
Python	3.12.3	Runtime for analyses and figure builders
`fastp`	1.3.3	FASTQ trimming and QC
`bowtie2`	2.5.5	hg38 read alignment
`samtools`/htslib	1.23.1/1.23.1	BAM filtering and QC
`bedtools`	2.31.1	Region sorting, merging, and blacklist subtraction
MACS3	3.0.4	ATAC-seq peak calling
NumPy/pandas/SciPy	2.4.6/2.3.3/1.17.1	Numerical and tabular analysis
matplotlib/pyBigWig	3.10.9/0.3.25	Figures and bigWig signal I/O
pysam/Biopython	0.23.3/1.87	BAM I/O and sequence utilities
JASPAR CORE vertebrates	2026 non-redundant	Known motif database
MEME Suite/STREME/Tomtom	5.5.9	De novo motif discovery and discovered-motif matching
Cell Ranger ARC	2.0.0	Source 10x PBMC Multiome processing
hg38 reference and blacklist	GRCh38/hg38; blacklist v2	Alignment, sequence extraction, and artifact filtering

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